- We learnt in the previous class that all living things, whether plants or animals reproduce.
- The process by which a living organism produces individuals of its own kind to continue its species is called **reproduction.**
- Let us study how reproduction occurs in plants.
- Flowering plants produce flowers that eventually develop into fruits and seeds.
- These seeds can grow into new plants.
- Because of this, flowers are known as the reproductive parts of a plant.
- However, a plant can also reproduce using its vegetative parts, such as roots, stems and leaves.

- **New plants can be created by just one parent or through the involvement of two parents — a male and a female.**
- **Based on how many parents are involved, there are two types of reproduction: Asexual and Sexual.**

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                Modes of Reproduction
                        |
                        |
        ┌───────────────┴───────────────┐
        |                               |
      Asexual                         Sexual
        |
        |
┌───────┴───────┬───────┬───────┐
|               |       |       |
Fragmentation  Spore formation  Budding  Vegetative propagation
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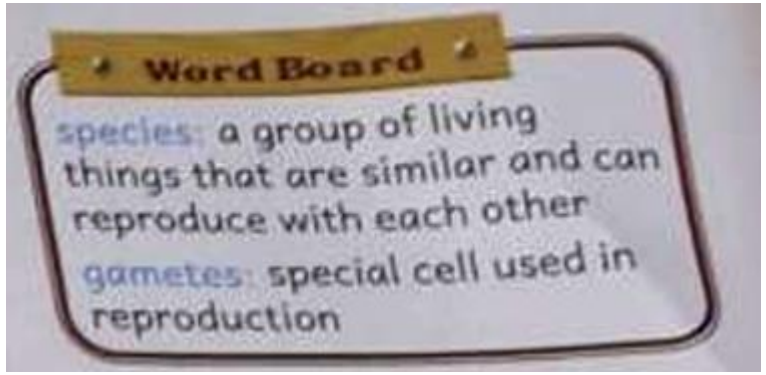
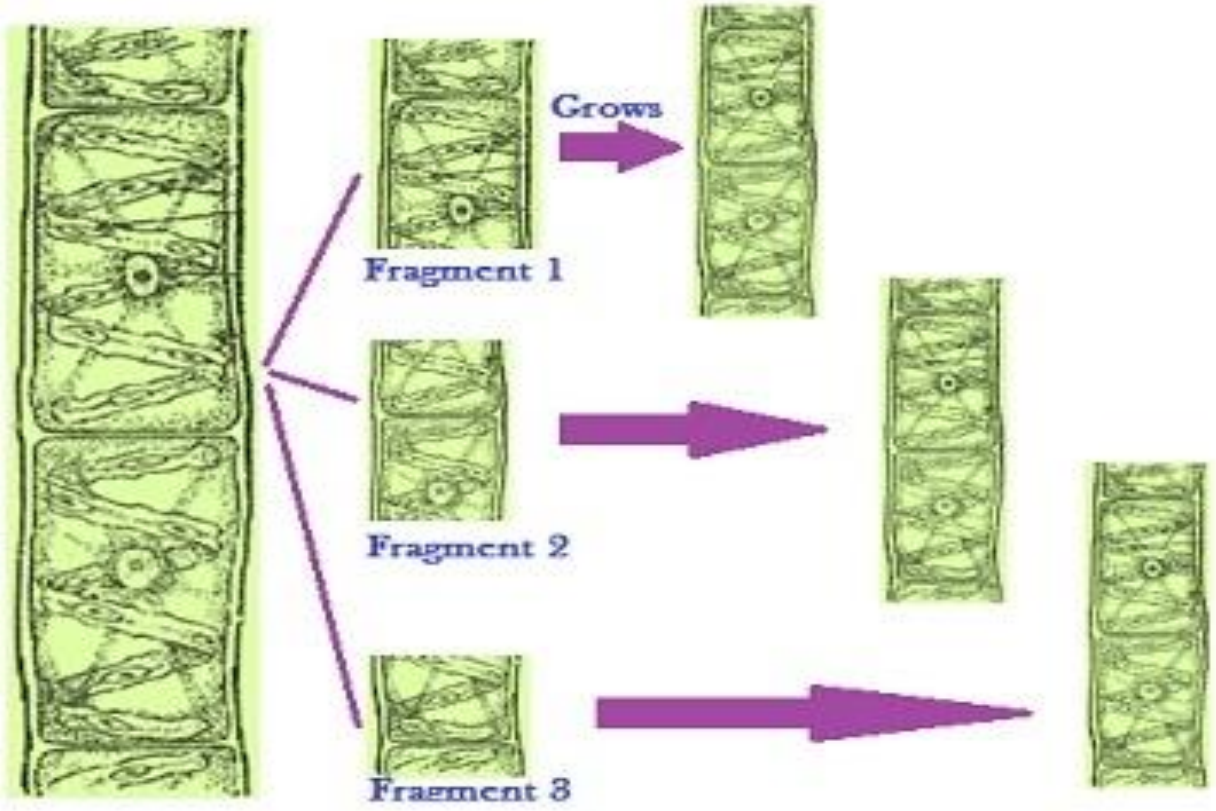


ASEXUAL REPRODUCTION

- **The method of reproduction which involves only a single parent without the formation of gametes or seeds is called asexual reproduction.**
- **It occurs in the vegetative parts of a plant.**
- **Since new plants arise from a single parent, the offspring are identical to the parent and to each other as well.**
- **Plants reproduce asexually by fragmentation, budding, spore formation, and vegetative propagation.**

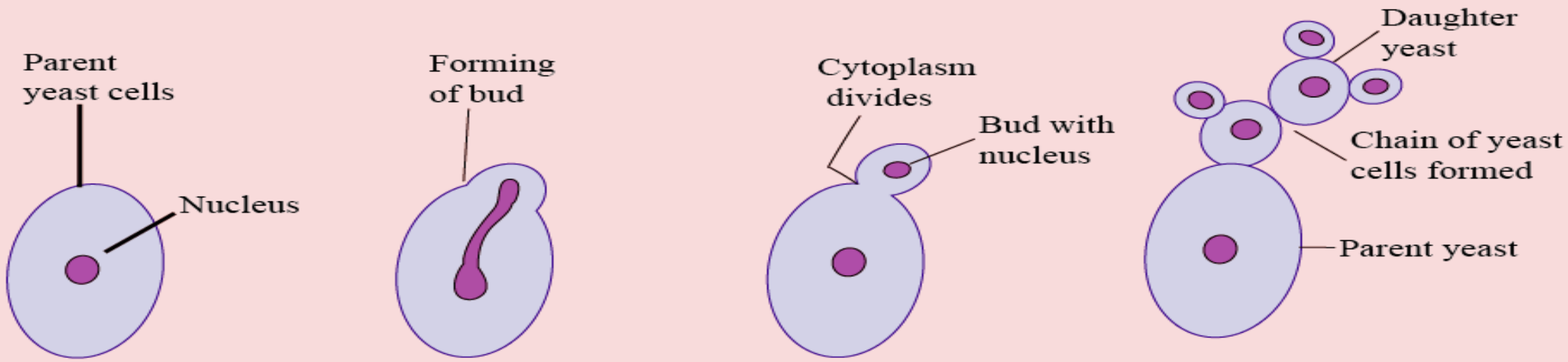
Fragmentation

- Fragmentation is a common method of asexual reproduction in filamentous algae like Spirogyra.
- When water and nutrients are present, the alga's filament splits into two or more parts, called fragments, and each fragment develops into a new individual.



Budding

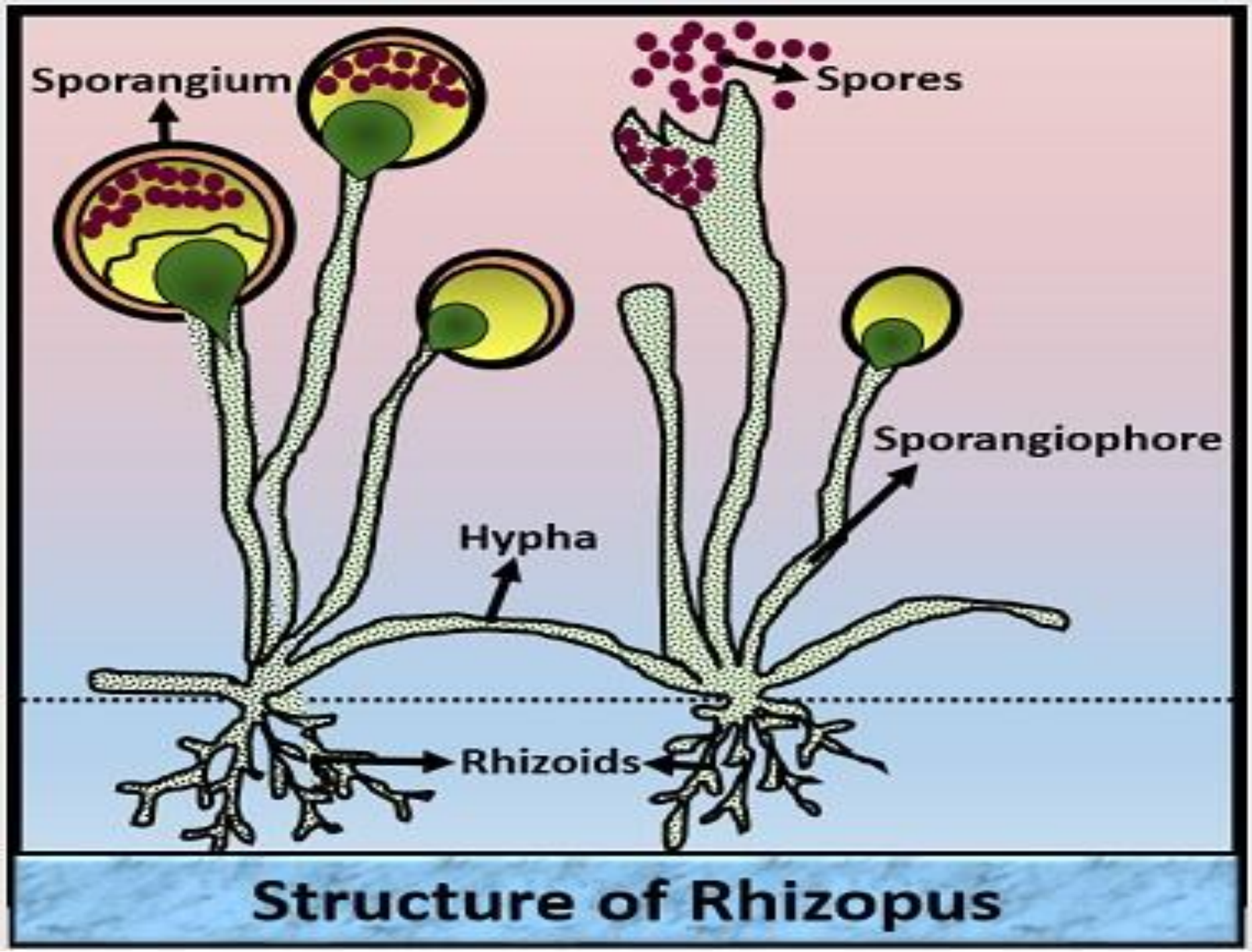
- In this process, a small bump or outgrowth called a bud forms on the body of the organism.
- The bud grows bigger and eventually separates from the parent body to become a new individual.
- Sometimes, a new bud starts forming from the first bud before it detaches, creating a chain of buds.
- Budding is commonly seen in yeast cells.



Budding in yeast

Spore formation

- Plants such as ferns and mosses bear asexual bodies called spores, which are contained in special structures called sporangia (singular: sporangium).
- These sporangia burst open to release spores.
- Each spore is covered by a hard outer covering which helps it withstand harsh conditions like high temperature and lack of moisture.
- Upon restoration of favourable conditions of the atmosphere and availability of food, each spore germinates into a new plant.
- Apart from ferns and mosses, a fungus called bread mould (*Rhizopus*), also reproduces by spore formation.
- They contain long thread-like structures called hyphae, each terminating into a bulb-like structure called sporangium, which bears spores.


Sporangium
Spores
Sporangiophore
Hypha
Rhizoids
Structure of Rhizopus

Artificial vegetative propagation is a technique where a part of the plant (stem, root, or leaf) is cut and planted in the soil to produce new plants. Cutting, layering, grafting, suckering, and tissue culturing are the methods used for artificial vegetative propagation.

Vegetative propagation

- Some plants can grow from other parts of the plant besides seeds.
- The process of growing a plant from other parts of a plant is known as vegetative propagation.
- The part of the plant from which the new plant grows is known as the vegetative part.
- Natural vegetative propagation occurs by a plant's roots, stems, or leaves.

Vegetative propagation by stem

- Vegetative propagation by stems can be seen from tubers (potato), bulbs (onion), and rhizomes (ginger).

Tubers

- Underground stems of some plants store food and become swollen. They contain buds in a depression called eye, such as in potatoes. If a potato is cut into several pieces, each containing an eye, each piece will grow into a new plant.

Bulbs

- Underground stems of some plants are covered by scaly leaves.
- These leaves store food and become thick and fleshy.
- They form a structure called bulb.
- It is found in onions, tulips, gladiolus, and lilies.

Rhizomes

- Underground stems of plants like ginger, turmeric, and banana have scaly leaves and buds on them.
- The buds develop outgrowths which can give rise to new plants.

DAY 1

Vegetative propagation by root

- Plants like carrots, radishes, and sweet potatoes reproduce from their roots.
- These plants store food in their roots.
- A new plant can grow when a portion of the root containing a vegetative bud is planted in the soil.

Vegetative propagation by leaf

- Plants such as *Bryophyllum* and Begonia reproduce from their leaves.
- They have buds on the margins of the leaves.
- Small plants grow from these buds.
- When they fall on the ground, they grow into new plants.

1. **Reproduction is the process by which living organisms produce**

   a) Food

   b) Energy

   c) Individuals of their own kind

   d) Oxygen

c

2. **Flowers are called reproductive parts of plants because they**

   a) Absorb water

   b) Produce seeds and fruits

   c) Store food

   d) Protect roots

b

3. **New plants produced from seeds show reproduction through**

   a) Vegetative parts only

   b) Roots only

   c) Sexual reproduction

   d) Fragmentation only

c

4. Reproduction involving only one parent is called

a) Sexual reproduction

b) Asexual reproduction

c) Cross reproduction

d) Hybrid reproduction

b

5. In asexual reproduction, the offspring are usually

a) Different from parents

b) Identical to parents

c) Larger than parents

d) Sterile

b

6. Fragmentation commonly occurs in which organism?

a) Moss

b) Spirogyra

c) Fern

d) Rose

b

7. **In fragmentation, each fragment develops into**

a) A leaf

b) A flower

c) A new individual

d) A root hair

c

8. **Budding is commonly seen in**

a) Yeast

b) Mango plant

c) Fern plant

d) Rose plant

a

9. **In budding, the new organism develops from**

a) Seeds

b) Spores

c) Bud or outgrowth

d) Roots

c

10. Spores are commonly produced by plants like

a) Mango and neem

b) Ferns and mosses

c) Roses and sunflowers

d) Wheat and rice

b

11. The structure that contains spores is called

a) Stigma

b) Sporangium

c) Bud

d) Node

b

12. Bread mould (Rhizopus) reproduces mainly through

a) Seeds

b) Budding

c) Spores

d) Grafting

c

13. Vegetative propagation means growing new plants from

a) Seeds only

b) Flowers only

c) Plant parts like stem, root, or leaf

d) Fruits only

c

14. Potato is an example of vegetative propagation through

a) Leaf

b) Flower

c) Tuber (stem)

d) Root hair

c

15. Onion is an example of vegetative propagation through

a) Bulb

b) Rhizome

c) Tuber

d) Spore

a

16. Ginger reproduces through

a) Bulb

b) Rhizome

c) Tuber

d) Spore

b

17. Plants like carrot and radish reproduce through

a) Stem

b) Flower

c) Root

d) Leaf

c

18. Bryophyllum produces new plants from

a) Roots

b) Flowers

c) Leaf margins

d) Seeds only

c

19. Artificial vegetative propagation includes methods like

a) Fragmentation and budding

b) Cutting, grafting, and layering

c) Pollination and fertilization

d) Spore bursting

b